

Fundamental Trigonometric Identities

ANSWER KEY



Using the Pythagorean identities

- 1 Given that $\sin \theta = \frac{3}{5}$ and θ lies in Quadrant II, find:

a $\cos \theta = -\frac{4}{5}$ **b** $\tan \theta = -\frac{3}{4}$ **c** $\sec \theta = -\frac{5}{4}$ **d** $\csc \theta = \frac{5}{3}$

- 2 Given that $\tan \theta = 2$ and θ lies in Quadrant III, find $\sin \theta$ and $\cos \theta$ in exact simplified form.

$\sec^2 \theta = 1 + \tan^2 \theta = 5$, and in QIII $\cos \theta < 0$, so $\cos \theta = -\frac{\sqrt{5}}{5}$ and $\sin \theta = \tan \theta \cos \theta = -\frac{2\sqrt{5}}{5}$.

Simplifying and verifying

- 3 Simplify each expression to a single trigonometric function:

a $\frac{1 - \cos^2 x}{\sin x} = \frac{\sin^2 x}{\sin x} = \sin x$

b $\sec x \cot x = \frac{1}{\cos x} \cdot \frac{\cos x}{\sin x} = \csc x$

c $\tan x \csc x = \frac{\sin x}{\cos x} \cdot \frac{1}{\sin x} = \sec x$

- 4 Verify the identity $(1 + \tan^2 x) \cos^2 x = 1$.

$(1 + \tan^2 x) \cos^2 x = \sec^2 x \cos^2 x = \frac{1}{\cos^2 x} \cos^2 x = 1$.

- 5 Verify the identity $\frac{\sin x}{1 + \cos x} = \frac{1 - \cos x}{\sin x}$.

Cross-multiplying view: $\sin^2 x = (1 + \cos x)(1 - \cos x) = 1 - \cos^2 x$, which is the Pythagorean identity.

Equivalently, multiply the left side by $\frac{1 - \cos x}{1 - \cos x}$ to obtain $\frac{\sin x(1 - \cos x)}{1 - \cos^2 x} = \frac{\sin x(1 - \cos x)}{\sin^2 x} = \frac{1 - \cos x}{\sin x}$.

Quotient and reciprocal identities

- 6 Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

a $\cot x \sin x = \frac{\cos x}{\sin x} \cdot \sin x = \cos x$

b $\frac{\sec x}{\csc x} = \frac{1/\cos x}{1/\sin x} = \frac{\sin x}{\cos x} = \tan x$

c $1 - (\sin^2 x \csc^2 x) \sin^2 x = 1 - \sin^2 x = \cos^2 x$ (since $\csc^2 x \sin^2 x = 1$)

- 7 Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.

Combine over $\cos x(1 - \sin x)$: $\frac{\cos^2 x - \sin x(1 - \sin x)}{\cos x(1 - \sin x)} = \frac{\cos^2 x + \sin^2 x - \sin x}{\cos x(1 - \sin x)} = \frac{1 - \sin x}{\cos x(1 - \sin x)} = \sec x$.

More identity verifications

- 8 Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.
The identity $1 + \tan^2 x = \sec^2 x$ (dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$) rearranges directly to $\sec^2 x - \tan^2 x = 1$.
- 9 Verify the identity $\frac{1}{1 - \cos x} + \frac{1}{1 + \cos x} = 2\csc^2 x$.
Combine over a common denominator: $\frac{(1 + \cos x) + (1 - \cos x)}{(1 - \cos x)(1 + \cos x)} = \frac{2}{1 - \cos^2 x} = \frac{2}{\sin^2 x} = 2\csc^2 x$.
- 10 Show that $(\sin x + \cos x)^2 = 1 + 2\sin x \cos x$.
Expand:
 $(\sin x + \cos x)^2 = \sin^2 x + 2\sin x \cos x + \cos^2 x = (\sin^2 x + \cos^2 x) + 2\sin x \cos x = 1 + 2\sin x \cos x$.
- 11 Given that $\cos \theta = -\frac{8}{17}$ and θ lies in Quadrant III, find $\sin \theta$, $\tan \theta$, and $\cot \theta$ in exact simplified form.
 $\sin^2 \theta = 1 - \frac{64}{289} = \frac{225}{289}$, and in QIII $\sin \theta < 0$: $\sin \theta = -\frac{15}{17}$. $\tan \theta = \frac{-15/17}{-8/17} = \frac{15}{8}$. $\cot \theta = \frac{8}{15}$.

Identities with sums and products

- 12 Verify the identity $\sin x \cos x (\tan x + \cot x) = 1$.
 $\sin x \cos x \left(\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} \right) = \sin^2 x + \cos^2 x = 1$.
- 13 Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.
This is a difference of squares: $\sec^2 x - \tan^2 x = 1$ (a Pythagorean identity).

Identities involving all three Pythagorean forms

- 14 State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).
 $\sin^2 x + \cos^2 x = 1$; $1 + \tan^2 x = \sec^2 x$; $1 + \cot^2 x = \csc^2 x$.
- 15 Given $\cot \theta = \frac{5}{12}$ and θ in Quadrant I, find $\csc \theta$ and $\sin \theta$ using the cotangent-cosecant identity.
 $\csc^2 \theta = 1 + \frac{25}{144} = \frac{169}{144}$, and in QI $\csc \theta > 0$: $\csc \theta = \frac{13}{12}$. $\sin \theta = \frac{1}{\csc \theta} = \frac{12}{13}$.

Simplifying more complex expressions

- 16 Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.
Factor as a sum of cubes: $\sin^3 x + \cos^3 x = (\sin x + \cos x)(\sin^2 x - \sin x \cos x + \cos^2 x)$. Dividing:
 $\frac{(\sin x + \cos x)(1 - \sin x \cos x)}{\sin x + \cos x} = 1 - \sin x \cos x$.

Verifying identities with a common denominator strategy

- 17 Verify $\tan x + \cot x = \sec x \csc x$.
 $\tan x + \cot x = \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{\sin^2 x + \cos^2 x}{\sin x \cos x} = \frac{1}{\sin x \cos x} = \sec x \csc x$.

- 18 Verify $\frac{\sin x}{\csc x} + \frac{\cos x}{\sec x} = 1$.

$$\frac{\sin x}{\csc x} = \sin x \cdot \sin x = \sin^2 x; \frac{\cos x}{\sec x} = \cos x \cdot \cos x = \cos^2 x. \text{ Sum: } \sin^2 x + \cos^2 x = 1.$$

Numeric identity verification (checking a specific angle)

- 19 Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

$$\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}: \text{LHS} = 1 + \frac{1}{3} = \frac{4}{3}. \sec \frac{\pi}{6} = \frac{1}{\cos(\pi/6)} = \frac{2}{\sqrt{3}}: \text{RHS} = \frac{4}{3}. \text{ Both sides match, confirming the identity at this angle (though this alone doesn't prove it for all } x).$$

Identities with squared expressions

- 20 Simplify $(1 - \sin^2 x)(1 + \tan^2 x)$ to a constant.

$$(1 - \sin^2 x) = \cos^2 x \text{ and } (1 + \tan^2 x) = \sec^2 x = \frac{1}{\cos^2 x}. \text{ Product: } \cos^2 x \cdot \frac{1}{\cos^2 x} = 1.$$

- 21 Given that $\sec \theta = -\frac{13}{5}$ and θ lies in Quadrant II, use the Pythagorean identity $1 + \tan^2 \theta = \sec^2 \theta$ to find $\tan \theta$ exactly, and then find $\sin \theta$ using the quotient identity.

$$\tan^2 \theta = \sec^2 \theta - 1 = \frac{169}{25} - 1 = \frac{144}{25}, \text{ and in QII } \tan \theta < 0: \tan \theta = -\frac{12}{5}. \text{ Since } \cos \theta = \frac{1}{\sec \theta} = -\frac{5}{13}: \\ \sin \theta = \tan \theta \cos \theta = \left(-\frac{12}{5}\right)\left(-\frac{5}{13}\right) = \frac{12}{13}.$$

- 22 Verify the identity $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ by cross-multiplying and using the Pythagorean identity.

Cross-multiplying: $(\csc x - 1)(\csc x + 1) = \cot^2 x$, i.e. $\csc^2 x - 1 = \cot^2 x$, which is exactly the Pythagorean identity $1 + \cot^2 x = \csc^2 x$ rearranged — confirming the identity holds.